ENGINEERING & ARCHITECTURE MASTER BLUEPRINT

PavePort Platform

Full-Scale Construction Commerce, Multi-Tier Commercial Quotations, Real-Time Yard ERP Inventory & Fleet Haulage Logistics Platform

Platform Scope

4 integrated product experiences (Super Admin Portal, Business Owner Yard ERP, Customer Commerce App, Delivery Partner Terminal) engineered on a unified domain model and real-time state machine.

Production Target Stack

Go 1.24+ REST API under /api/v1, PostgreSQL hosted on Neon, Redis 7 caching & distributed locks, Expo React Native mobile apps, and Cloudflare + AWS infrastructure.

Delivery Roadmap

Comprehensive 2-Month (8-Week) sprint-by-sprint development, integration, security audit, and testing execution plan leading to pilot deployment.

Security & Governance

Zero-Trust RBAC governance (10 User Roles, 23 Granular Permissions), tenant isolation, cryptographic delivery OTP handshake, and immutable audit trails.

Prepared By: Principal Software Architect & QA Lead

Target Release: Version 2.0 Production Pilot

Document Date: October 2026

Classification: Confidential / Client Deliverable

1.0 Executive Summary & Industry Problem Solution

1.1 Market Reality & Industry Problem

The construction materials industry ($1.2T+ globally, $150B+ in South Asia) remains one of the most operationally fragmented commercial sectors. Civil contractors, builders, and individual homebuilders face severe friction during material procurement, while material dealers and logistics operators operate on pen-and-paper systems:

The Procurement Challenge (Customer / Builder)

- **Opaque, Volatile Pricing:** Material prices (cement, TMT steel, M-sand) fluctuate daily without centralized price discovery or grade transparency.
- **Unreliable Inventory Commitments:** Suppliers accept orders only to disclose out-of-stock grades hours later, halting concrete pours.
- **Manual, Slow RFQ Cycles:** Requesting bulk commercial bids requires phone calls, paper slips, and days of negotiation.
- **Transit & Unloading Disputes:** Disagreements regarding drop-off arrival times, weighbridge discrepancy, and damaged bags.

The Supplier & Logistics Challenge (Yard / Driver)

- **No Real-Time Yard ERP:** Suppliers lack automated stock reservation when orders arrive, leading to double-allocation.
- **Statutory & KYC Risks:** Operating without digital verification of GSTIN, trade licenses, or weighbridge calibration certificates.
- **Idle Logistics Capacity:** Heavy tippers and mini-trucks (Tata Ace, 10-wheelers) return empty due to uncoordinated dispatch.
- **Cash Leakage:** Reliance on unverified informal credit notes with payment recovery cycles extending past 60-90 days.

1.2 The PavePort Solution Architecture

PavePort solves this friction through an integrated 4-product ecosystem built on a shared domain model, guaranteed stock allocation, transparent bidding, and verifiable logistics execution:

DOMAIN PROBLEM	TRADITIONAL MECHANISM	PAVEPORT ENGINEERED SOLUTION
Material Discovery	Local phone inquiries, unverified middlemen, unknown stock levels.	**Location-Aware Marketplace:** Pincode radius routing, real-time yard catalog with BIS/IS certified grades.
Commercial Pricing & Bids	Unwritten verbal promises prone to billing discrepancies.	**Formal Bidding Engine:** Formal digital quotation workflow with validity period, itemized transport, GST breakdown, and one-tap order conversion.
Inventory Integrity	Manual warehouse registers updated at end of week.	**Strict Stock Reservation State Machine:** Material stock is automatically reserved when an order is placed, preventing overselling.
Logistics Execution	Driver phone calls, unmonitored routes, delivery disputes.	**Automated Dispatch & OTP Handshake:** Instant fleet broadcast to nearby trucks, corridor telemetry, and 4-digit site OTP verification.
Platform Governance	Zero centralized visibility into vendor compliance.	**Centralized Admin & Granular RBAC:** 10 distinct roles, 23 permissions, vendor KYC audits, and immutable event logs.

2.0 One Platform, Four Connected Product Suites

Rather than isolated applications, PavePort operates as one integrated platform with tailored interfaces suited to each participant's physical environment.

1. Admin Portal (Desktop Web)

`Super / Ops Admin`

Centralized operations console designed for desktop workstations with high information density:

- **KYC & Store Approvals:** Verification of GSTIN, trade licenses, yard physical addresses, and geofence coverage.
- **Catalog & Category Moderation:** Standardization of units of measurement (Bags, Metric Tonnes, Cubic Ft).
- **Order & Fulfillment Oversight:** Real-time platform-wide order lifecycle tracking and dispute mediation.
- **Governance & Audit Trails:** Granular RBAC privilege assignment and immutable event logging.

2. Business Owner App (Handheld Yard ERP)

`Supplier Yard Owner`

Handheld mobile app engineered for yard managers moving between warehouse bays and weighbridges:

- **Live Yard Ledger:** Stock ledger with instant inflow, reserved, and dispatched quantity counters.
- **Quotation Formulation:** Respond to contractor RFQs with quoted unit rates, freight, and tax in under 60 seconds.
- **Order Staging Bay:** Accept incoming orders, trigger pallet loading, and mark loads "Ready for Dispatch".
- **Carrier Check-In:** View assigned driver details, vehicle registration, and verify outbound load manifests.

3. Customer App (Procurement Marketplace)

`Contractor / Builder`

Mobile-first procurement app designed for site engineers and civil contractors on active job sites:

- **Geo-Located Yard Discovery:** Filter verified yards by pincode, delivery window, and material certifications.
- **Bulk Estimate RFQ:** Submit commercial quote requests with site unloading notes for 50-500+ bags.
- **Contractor Procurement Cart:** Single-supplier cart staging with dynamic transport and statutory GST calculations.
- **Live 8-Stage Timeline & OTP:** Real-time order progress from yard loading to transit, showing secure site OTP (4829).

4. Delivery Partner App (In-Cab Haulage Terminal)

`Fleet Carrier / Driver`

High-contrast, large-touch terminal tailored for drivers operating commercial haulage trucks:

- **Available Jobs Board:** Live listing of staged yard loads with guaranteed compensation payouts (e.g., ₹1,200).
- **Corridor Navigation:** Pickup yard loading bay address, transit corridor, and site destination details.
- **Two-Tap Milestone Telemetry:** Confirm Yard Pickup → Start Transit to Job Site.
- **Cryptographic OTP Verification:** Driver enters contractor's 4-digit code to execute cryptographically verified drop-off.

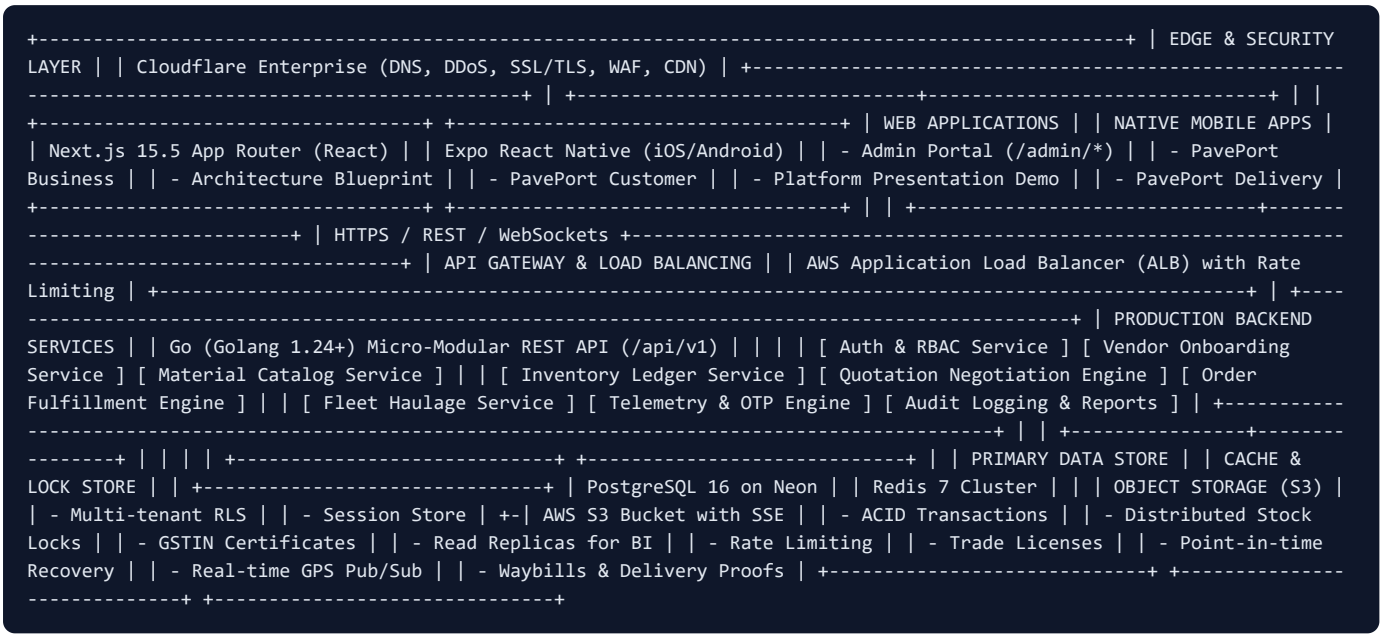
2.1 Role-Based Access Control (RBAC) Security Architecture

PavePort enforces strict privilege segregation across 10 user personas and 23 fine-grained permissions:

ROLE IDENTIFIER	TARGET PERSONA	PERMITTED OPERATIONS	SYSTEM BOUNDARIES
SUPER_ADMIN	Platform Executives & Chief Architect	All 23 permissions (Full create, read, update, delete, system config, RBAC)	Universal Platform Access
OPERATIONS_ADMIN	Regional Hub Operations Directors	Order oversight, business approvals, logistics assignment, dispute overrides	Cross-Business Operations
BUSINESS_VERIFICATION_ADMIN	Statutory Compliance Officers	Review GSTIN, Trade Licenses, weighbridge certs; approve/reject vendor onboarding	Vendor Compliance Scope
CATALOG_MANAGER	Material Specification Experts	Standardize BIS grades (OPC 53, Fe 550D), create categories, approve listings	Product Catalog Scope
LOGISTICS_MANAGER	Fleet Dispatch Coordinators	Monitor haulage assignments, driver compliance, route exceptions	Logistics & Transit Scope
BUSINESS_OWNER	Supplier Yard Owners & Managers	Manage own yard catalog, stock levels, submit formal quotes, fulfill orders	Strict Single-Tenant Isolation
CUSTOMER	Civil Builders & Site Contractors	Browse materials, request quotations, execute purchase orders, sign off via OTP	Customer Account Scope
DELIVERY_PARTNER	Commercial Haulage Truck Drivers	Claim available loads, report milestone telemetry, submit OTP to clear payout	Assigned Trip Scope

3.0 System Architecture & Target Production Technology Stack

3.1 Production Architecture Blueprint (C4 Container View)



3.2 Technical Stack Specification

LAYER	TECHNOLOGY CHOICE	ARCHITECTURAL RATIONALE & IMPLEMENTATION DETAILS
Backend API	**Go (Golang 1.24+)**	Sub-millisecond execution overhead, lightweight goroutines for concurrent truck GPS telemetry, static binary deployments, modular domain-driven layout without heavy ORM bloat.
Database	**PostgreSQL 16 (Neon)**	Serverless auto-scaling, strict relational ACID compliance for financial orders and stock balances, Row-Level Security (RLS) for tenant isolation, automated branching for testing.
Caching & Telemetry	**Redis 7.x Cluster**	Distributed locking for stock reservations (prevents race conditions when 2 contractors claim identical cement bags simultaneously); sub-10ms pub/sub for driver location updates.
Prototype Frontend	**Next.js 15.5 App Router**	React 19, TypeScript strict mode, Tailwind CSS for industrial UI tokens, Three.js/R3F for interactive 3D hero construction scene, Zustand for shared cross-product state.
Future Native Apps	**Expo React Native**	Universal code-sharing across iOS and Android, NativeWind styling, offline SQLite caching for erratic yard network connectivity, camera module for delivery invoice scanning.
DevOps & Edge	**Cloudflare + Docker + AWS**	Cloudflare WAF with OWASP rulesets; multi-stage Docker containers deployed on AWS ECS/Fargate; GitHub Actions automated lint, test, build, and deployment pipelines.

4.0 Lifecycle State Machines & End-to-End User Flows

4.1 The 9-Stage Milestone Master Workflow

The platform guarantees end-to-end operational integrity through a continuous workflow connecting the 4 applications:

STEP	APPLICATION	ACTOR / PERSONA	OPERATIONAL ACTION & TECHNICAL TRANSITION
1	Admin Portal	Super Admin (All Access)	Reviews multi-tenant platform dashboard, KYC queue, and confirms system health metrics.
2	Admin Portal	Verification Admin	Inspects vendor *Sri Lakshmi Cement*, verifies GSTIN & BBMP trade licenses, approves store.
3	Customer App	Ramesh Kumar (InfraBuild)	Locates Sri Lakshmi Cement via pincode 560038, configures 100 bags UltraTech Cement, submits RFQ.
4	Business App	Venkatesh Murthy (Owner)	Formulates official bid: ₹380/bag, ₹1,200 transport, ₹6,840 GST, valid 7 days. Submits to customer.
5	Customer App	Ramesh Kumar (InfraBuild)	Reviews binding quotation (₹46,040 total). Clicks *Accept Quotation* → generates Order PV-ORD-2026-9041.
6	Business App	Venkatesh Murthy (Owner)	Accepts PO, locks 100 bags from available to reserved stock, wraps pallets, clicks *Mark Ready for Dispatch*.
7	Delivery App	Murugan Selvam (Tata Ace)	Inspects available corridor jobs. Claims load with guaranteed ₹1,200 payout. Status → *ASSIGNED*.
8	Delivery App	Murugan Selvam (Tata Ace)	Confirms yard pickup → starts transit → arrives at site. Enters customer OTP **4829** to complete drop-off.
9	Admin Portal	Super Admin	Inspects final order completion, ₹1,200 driver wallet credit, 100-bag inventory deduction, and audit logs.

4.2 Mathematical Quotation & Pricing Engine

The platform executes deterministic financial arithmetic with zero rounding leakage:

```
Subtotal (S) = Sum(Item_Quantity_i * Quoted_Unit_Price_i) Delivery Freight (D) = Base_Haulage_Rate + (Distance_Km * Per_Km_Tonnage_Factor) Statutory GST (T) = (Subtotal + Delivery_Freight) * 0.18 (Standard Central + State GST) Grand Total (G) = S + D + T Stock Transition on Acceptance: Available_Stock = Available_Stock - Requested_Quantity Reserved_Stock = Reserved_Stock + Requested_Quantity Stock Transition on Dispatch: Reserved_Stock = Reserved_Stock - Requested_Quantity Dispatched_Stock = Dispatched_Stock + Requested_Quantity
```

4.3 Cryptographic OTP Delivery Verification

Delivery disputes are eliminated via a client-generated 4-digit verification code. When Order `PV-ORD-2026-9041` is created, the system securely generates OTP `4829` visible only to the customer on their live order timeline. The haulage driver cannot sign off delivery without physically standing at the site and entering this code into the driver terminal, triggering instant compensation clearing to the driver wallet.

5.0 Database Schema & Relational Entity Model

The target PostgreSQL database implements strict relational integrity and foreign key constraints:

TABLE NAME	PRIMARY FIELDS & DATA TYPES	RELATIONAL CONSTRAINTS & INDEXES
users	`id (UUID PK), email, phone, role (ENUM), password_hash, is_active, created_at`	Unique index on email and phone. Indexed by role for RBAC filtering.
businesses	`id (UUID PK), name, slug, gstin, owner_id (FK), status (ENUM), min_order, rating`	FK to users(id). Unique index on gstin and slug. B-tree index on status.
business_documents	`id (UUID PK), business_id (FK), doc_type, file_url, doc_number, status, verified_at`	FK to businesses(id) ON DELETE CASCADE. Indexed on business_id.
products	`id (UUID PK), business_id (FK), category_id (FK), name, brand, grade, price, unit, stock`	FK to businesses(id), categories(id). Compound index on (business_id, category_id).
inventory_movements	`id (UUID PK), product_id (FK), business_id (FK), delta, prev_stock, new_stock, type, reason`	Append-only ledger. FK to products(id), businesses(id). Indexed on created_at.
quotations	`id (UUID PK), quote_number, customer_id (FK), business_id (FK), subtotal, tax, status, expires_at`	FK to users(id), businesses(id). Unique quote_number. Indexed on status.
quotation_items	`id (UUID PK), quotation_id (FK), product_id (FK), qty, quoted_rate, subtotal`	FK to quotations(id) ON DELETE CASCADE, products(id).
orders	`id (UUID PK), order_number, customer_id (FK), business_id (FK), total, status, delivery_otp`	FK to users(id), businesses(id). Unique order_number. Indexed on (customer_id, status).
delivery_assignments	`id (UUID PK), order_id (FK), driver_id (FK), status, pickup_geo, dropoff_geo, payout`	FK to orders(id), users(id). Indexed on (driver_id, status) and order_id.
audit_logs	`id (UUID PK), actor_id (FK), action, entity_type, entity_id, ip_address, timestamp`	Immutable partition by month. Indexed on (actor_id, timestamp) and entity_id.

6.0 Comprehensive 2-Month (8-Week) Development & Testing Roadmap

This sprint schedule defines the step-by-step production implementation across 4 two-week agile sprints. It covers backend engineering, mobile app development, automated testing gates, and production go-live:

> **Timeline Parameters & Milestones**
>
> **Total Duration:** 8 Weeks (60 Calendar Days) • **Engineers:** 1 Principal Architect / Backend Lead, 2 Full-Stack/Mobile Engineers, 1 QA Automation Engineer • **Sprint Rhythm:** 2-Week Sprints with Bi-Weekly Functional Client Demos.

6.1 Sprint-by-Sprint Execution Plan

SPRINT / WEEK	BACKEND & INFRASTRUCTURE WORKSTREAM	WEB & MOBILE WORKSTREAM	TESTING, QA & SECURITY GATES
Sprint 1 (Weeks 1-2) `Foundation`	• Setup Go REST API modular project layout. • Deploy Neon PostgreSQL & Redis cluster. • Write database schema migrations (Goose). • Implement JWT authentication & RBAC middleware.	• Initialize Next.js 15.5 production repo. • Initialize Expo React Native workspace. • Configure Tailwind & NativeWind design system. • Implement persistent user persona switcher.	**Gate 1 Testing:** • Unit tests for RBAC permission evaluator. • Database migration integrity & rollback tests. • Automated CI/CD pipeline on GitHub Actions. • API contract tests with Swagger/OpenAPI.
Sprint 2 (Weeks 3-4) `Commerce & ERP`	• Vendor onboarding & document upload to S3. • Material catalog CRUD & BIS grade tagging. • Inventory movement ledger & stock decrement. • Commercial RFQ submission & bidding endpoints.	• **Business App:** Mobile stock ledger & inventory adjustment stepper. • **Business App:** Bid formulation screen. • **Customer App:** Pincode routing & catalog. • **Customer App:** RFQ request sheet.	**Gate 2 Testing:** • Stock race-condition tests (Redis locks). • Unit tests for GST and freight pricing math. • Mobile responsive checks across iPhone & Android. • Mock S3 document upload integration tests.
Sprint 3 (Weeks 5-6) `Orders & Fleet`	• Order lifecycle finite state machine. • Payment gateway integration (Razorpay / UPI). • Delivery job broadcast & driver claiming logic. • Telemetry webhook & cryptographic OTP handler.	• **Customer App:** Cart & Simulated Checkout. • **Customer App:** Live 8-stage order tracker. • **Delivery App:** Available loads terminal. • **Delivery App:** OTP verification modal.	**Gate 3 Testing:** • Full end-to-end Playwright order test flow. • OTP verification handshake security tests. • Driver payout calculation & ledger audit tests. • Network drop & offline reconnection handling.
Sprint 4 (Weeks 7-8) `Audit & Go-Live`	• Cloudflare WAF, TLS, and DDoS edge tuning. • Production AWS ECS/Fargate task definitions. • Immutable audit logging & compliance exports. • Automated database backup & DR runbooks.	• **Admin Portal:** Analytics dashboards (Recharts). • **Admin Portal:** Dispute resolution tools. • Native iOS TestFlight & Android APK builds. • Performance polish & asset minification.	**Gate 4 Final Acceptance:** • Static Application Security Testing (SAST). • Penetration testing & OWASP vulnerability scan. • k6 Load Testing (1,000 concurrent checkouts). • Pilot onboarding of 5 verified supply yards.

6.2 Quality Assurance & Testing Architecture

1. Automated Unit & Math Tests

Continuous testing of pricing formulas, GST percentage calculations, inventory reservation locks, and permission trees with 90%+ code coverage.

2. Playwright End-to-End Tests

Automated browser test suites that validate the complete 9-stage sequence: RFQ submission → Quote formulated → PO generated → Driver claimed → OTP verified.

3. Security SAST & Pen-Testing

Static code scanning for injection flaws, tenant data leakage, JWT expiration audits, strict CORS restrictions, and rate-limiting enforcement on all public routes.

7.0 DevOps, Cloud Infrastructure & Compliance

7.1 Cloudflare Edge & AWS Production Architecture

PavePort utilizes an industry-standard defense-in-depth model separating edge protection, application compute, and data persistence:

- **Cloudflare Edge:** Managed DNS, TLS termination, WAF with OWASP Core Rulesets, bot management against catalog scrapers, and edge asset caching.
- **AWS Compute (ECS / Fargate):** Containerized Go backend services running inside private subnets across multiple Availability Zones with auto-scaling based on CPU/memory load.
- **Neon PostgreSQL:** Serverless high-availability database cluster with automated multi-zone replication, automated daily backups, and point-in-time recovery (PITR).
- **AWS S3 Object Storage:** Private S3 buckets with Server-Side Encryption (AES-256) storing vendor GSTIN certificates, trade licenses, and signed waybill receipts. Pre-signed URLs with 15-minute expirations are used for secure client downloads.

7.2 Production Disaster Recovery & SLA Objectives

METRIC / TARGET	PRODUCTION SLA TARGET	OPERATIONAL MITIGATION STRATEGY
Recovery Point Objective (RPO)	≤ 5 Minutes	Continuous PostgreSQL Write-Ahead Log (WAL) archiving to cloud storage.
Recovery Time Objective (RTO)	≤ 15 Minutes	Automated ECS task replacement and multi-AZ database failover.
API Uptime Availability	99.95% Monthly	Zero-downtime rolling deployments via AWS Application Load Balancer.
P95 API Response Latency	< 120 ms	Redis caching for read-heavy catalog endpoints and connection pooling via pgx.

PavePort Platform Engineering Specification • Confidential © 2026 PavePort Technologies Inc. All rights reserved.